Lecture 5

# Function limits and continuity

**Uses.**

Summary 1: The number $\sqrt{2}$ is irrational. Quantifier negation reverses the choices.
Summary 2: Functions have specified domains. The triangle and reverse-triangle inequalities control errors. Rational numbers are dense in $\mathbb{R}$. The Archimedean property makes $1/n$ arbitrarily small.
Summary 3: A sequence converges when every positive error tolerance holds eventually. Real sequence limits are unique and obey the algebraic laws.
Summary 4: Order passes to limits. The squeeze theorem proves a limit from two agreeing bounds.
Standing fact E: School domains, algebraic identities and radian trigonometric bounds are assumed.

**Objectives.** *A function limit controls values near an input rather than terms after an index. We formulate that control by a distance threshold and connect it with sequences. Continuity then compares a nearby limit with the function's actual value. The domain determines which inputs are tested.*

By the end of this lecture you will be able to:

- Prove or disprove a function limit using distances or sequences.
- Apply the limit laws and state the hypothesis needed for composition.
- Interpret one-sided limits and limits involving infinity.
- Test relative continuity and use the basic sine limit.

## 5.1 A limit near a domain point

Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. A *neighbourhood* of $c \in \mathbb{R}$ of radius $r > 0$ is the interval $(c-r, c+r)$. Its *deleted neighbourhood* excludes $c$. A point $c \in \mathbb{R}$ is a *cluster point* of $A$ if every deleted neighbourhood of $c$ contains a member of $A$. In symbols,

$$\forall r > 0 \quad \exists x \in A : \quad 0 < |x - c| < r.$$

The point need not belong to $A$. For example, zero is a cluster point of $(0, 1)$, while 2 is not. A point $c \in A$ is *isolated* if some neighbourhood contains no other member of $A$. For $A = \{0\} \cup [1, 2]$, the point 0 is isolated: radius $1/2$ excludes all other inputs. A deleted limit condition there would hold for every proposed value, because no inputs are tested. The cluster-point requirement prevents this failure of uniqueness.

**Definition 5.1** (Function limit)**.** Suppose $c$ is a cluster point of $A$. The notation $\lim_{x\to c} f(x) = L$ means, for $L \in \mathbb{R}$,

$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \forall x \in A: \quad 0 < |x - c| < \delta \implies |f(x) - L| < \varepsilon.$$

The tolerance $\varepsilon$ is requested first. The radius $\delta$ may depend on it, but must work for every allowed input in the deleted neighbourhood. The definition does not inspect $f(c)$, even when that value is defined.

**Example 5.2** (A direct distance estimate)**.** For $f(x) = x^2$ on $\mathbb{R}$, we prove $f(x) \to 1$ as $x \to 1$. If $0 < |x - 1| < 1$, then $|x + 1| \le |x - 1| + 2 < 3$. Hence

$$|x^2 - 1| = |x - 1||x + 1| < 3|x - 1|.$$

Given $\varepsilon > 0$, choose $\delta = \min\{1, \varepsilon/3\}$. Every $x$ with $0 < |x - 1| < \delta$ then has error below $\varepsilon$. The first bound controls a variable coefficient. The second supplies the requested output error.

Figure 5.1 depicts the same argument for one requested accuracy. The indicated $\delta$ is a sufficient choice; the definition does not require the largest possible radius.

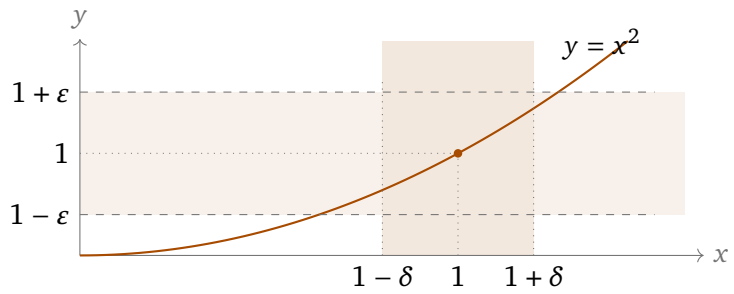



Figure 5.1: For $\varepsilon = 0.6$, the proof chooses $\delta = \min\{1, \varepsilon/3\} = 0.2$. Inputs in the deleted $\delta$-neighbourhood have outputs in the $\varepsilon$-band. The filled point $(1, 1)$ belongs to the graph even though the definition does not test the input $x = 1$.

**Theorem 5.3** (Sequential characterisation)**.** *At a cluster point $c$ of $A$, the limit $f(x) \to L$ holds if and only if for every sequence $x_n \in A \setminus \{c\}$ with $x_n \to c$, we have $f(x_n) \to L$.*

*Proof.* Suppose the distance definition holds. Given $\varepsilon > 0$, choose its $\delta$. Convergence $x_n \to c$ makes $|x_n - c| < \delta$ eventually. Since every $x_n \neq c$, the definition gives $|f(x_n) - L| < \varepsilon$ eventually, proving the image limit.

Conversely, if the distance definition fails, its negation supplies one $\varepsilon_0 > 0$ such that for every $\delta > 0$ some allowed input has output error at least $\varepsilon_0$. With $\delta = 1/n$, choose $x_n \in A$ satisfying

$$0 < |x_n - c| < 1/n, \qquad |f(x_n) - L| \ge \varepsilon_0.$$

The first estimate proves $x_n \to c$. The second prevents $f(x_n) \to L$. This contradicts the proposed sequential property and proves the converse. ∎

The cluster-point assumption supplies at least one such input sequence: choose its $n$th input in $A \setminus \{c\}$ within distance $1/n$. If a function had two limits, this one sequence of function values would converge to both. Sequence uniqueness from Lecture 3 therefore proves uniqueness of the function limit.

**Example 5.4** (Two approaches that disagree). Let $f(x) = \sin(1/x)$ for $x \neq 0$. The positive inputs

$$x_n = \frac{1}{\pi/2 + 2\pi n}, \qquad y_n = \frac{1}{3\pi/2 + 2\pi n}$$

both tend to zero: dividing each denominator by $n$ reduces this to Lecture 3's quotient law and $1/n \to 0$. Standing fact E gives $f(x_n) = 1$ and $f(y_n) = -1$. The sequential characterisation would force both image limits to equal any proposed function limit. Since they differ, the limit at zero does not exist. Assigning a value at zero cannot repair it.

## 5.2 Limit laws and composition

**Theorem 5.5** (Algebra, order and squeezing). *Suppose $f, g : A \to \mathbb{R}$ have limits $L, M$ at the same cluster point $c$, and let $\alpha \in \mathbb{R}$. Their sum, scalar multiple and product have limits $L + M$, $\alpha L$ and $LM$. If $M \neq 0$, then $g$ is nonzero in a deleted neighbourhood of $c$ and $f/g \to L/M$ there. If $f \leq g$ in a deleted neighbourhood, then $L \leq M$. For $h : A \to \mathbb{R}$, if $f \leq h \leq g$ there and $L = M$, then $h \to L$.*

*Proof.* Take any sequence of allowed inputs approaching $c$. The sequential characterisation gives sequence limits $L, M$. Lecture 3's algebraic laws and Lecture 4's order/squeeze results give each stated conclusion for the corresponding output sequences. Applying the characterisation in reverse proves the function statements. For the quotient, the nonzero denominator must also be checked on a whole neighbourhood. Choose a radius giving $|g(x) - M| < |M|/2$. The reverse triangle inequality then gives $|g(x)| > |M|/2 > 0$. The quotient is therefore defined on the needed tail of every approach sequence, as the proof requires. ∎

Constants tend to themselves: their output error is zero. The identity tends to the approach point, because its error is $|x - c|$.

For example, $f(x) = (x^2 - 1)/(x - 1)$ on $\mathbb{R} \setminus \{1\}$ agrees with $x + 1$. Thus $f(x) \to 2$ as $x \to 1$. Substitution into its original quotient would give a zero denominator and is not justified. Cancellation is valid on precisely the deleted domain used in the limit.

**Proposition 5.6** (Local boundedness and preservation of sign). *If $f(x) \to L \in \mathbb{R}$ as $x \to c$, then $f$ is bounded on some deleted neighbourhood of $c$. If $L > 0$, the values are positive there. If $L < 0$, they are negative there.*

*Proof.* The tolerance 1 gives a radius on which $|f(x)| \leq |f(x) - L| + |L| < 1 + |L|$. This proves local boundedness. For $L > 0$, tolerance $L/2$ gives $f(x) > L - L/2 = L/2 > 0$. For $L < 0$, tolerance $|L|/2$ gives $f(x) < L + |L|/2 = L/2 < 0$. The radii may differ. Each proves its stated claim. ∎

This is a local assertion. For example, $1/(x - 2)$ has limit $-1/2$ at zero, so is bounded near zero although it is unbounded near 2. Boundedness near a point alone does not imply a limit: $\sin(1/x)$ is bounded by 1 in absolute value and still has the two incompatible approaches already exhibited. A zero limit does not determine sign. For example, $-x^2 \to 0$ while its deleted values are negative.

**Theorem 5.7** (Composition with a deleted limit). *Suppose $g(x) \to b$ as $x \to c$, and $f(u) \to L$ as $u \to b$. Assume $g(x)$ belongs to the domain of $f$ for every allowed $x$ in some deleted neighbourhood of $c$. If either $g(x) \neq b$ in a deleted neighbourhood of $c$, or $b$ is in the domain of $f$ and $f(b) = L$, then $f(g(x)) \to L$.*

*Proof.* Given $\varepsilon > 0$, choose an outer radius $\eta > 0$ so that $0 < |u - b| < \eta$ gives $|f(u) - L| < \varepsilon$ for allowed $u$. The inner limit supplies a radius $\delta$ making $|g(x) - b| < \eta$. Decrease it to stay inside the neighbourhood where the composition is defined. Under the first alternative, decrease it also to the radius where $g(x) \neq b$. Under that alternative only the nonzero case below can occur. If $g(x) \neq b$, the outer estimate applies. If $g(x) = b$, the second hypothesis supplies $|f(g(x)) - L| = 0$. Thus every sufficiently near allowed input has output error below $\varepsilon$ in either case. ∎

The qualification matters. Set $g(x) = 0$ for all real $x$, and let $f(0) = 1$ while $f(u) = 0$ for $u \neq 0$. The inner limit is zero and the deleted outer limit at zero is zero, but $f(g(x))$ is constantly 1. Inner values equal to the excluded outer point must be handled, not silently discarded.

## 5.3 Directions and infinite limits

A one-sided limit tests only the specified side of the input. The point must be a cluster point from that side: every radius must contain a domain input strictly on the specified side. A limit at $+\infty$ requires a domain unbounded above. At $-\infty$ it requires one unbounded below. The following table gives the input conditions replacing the deleted condition at a finite point. The cutoff $\delta$ or $R$ is positive.

| Approach | Inputs tested in the domain |
|---|---|
| $x \to c$ | $0 < \lvert x - c \rvert < \delta$ |
| $x \to c+$ | $0 < x - c < \delta$ |
| $x \to c-$ | $0 < c - x < \delta$ |
| $x \to +\infty$ | $x > R$ |
| $x \to -\infty$ | $x < -R$ |

For a finite output limit $L$, every requested $\varepsilon > 0$ must admit one input cutoff such that all tested inputs satisfy $|f(x) - L| < \varepsilon$. For output $+\infty$, replace that requested tolerance by any $M > 0$ and require $f(x) > M$. For output $-\infty$, require $f(x) < -M$ instead. Infinity is not a real number to substitute into an algebraic limit law.

A two-sided limit persists on either restricted side having the point as a cluster point: its same radius still controls every tested input. Conversely, if both side limits exist and agree, choose the smaller of their radii for each requested error or bound. Every nonzero nearby input lies on one side, so this radius proves the common two-sided limit. This proves the one-sided criterion, including infinite output limits. Using positive bounds $M$ is equivalent to using every real bound: a positive bound larger than a requested real bound gives its test too.

The sequential characterisation also holds for each approach in the table. For a finite output limit, the forward proof uses the input sequence's cutoff for the chosen radius or large-input bound. If the limit fails, a fixed tolerance fails at an input for every cutoff. Choose such inputs within distance $1/n$ on the requested finite side, or beyond $n$ at $+\infty$, or below $-n$ at $-\infty$. They form an allowed approach sequence whose outputs fail that tolerance. For an infinite output limit, the same argument uses a fixed failed output bound instead of a tolerance. Consequently the finite-output arithmetic and order laws hold at these approaches too, by applying the sequence laws to every approach sequence. No law here treats infinity as a real output value.

For example, as $x \to 0+$ the function $1/x$ tends to $+\infty$. Given $M > 0$, choose $\delta = 1/M$. Then $0 < x < \delta$ implies $1/x > M$. As $x \to 0-$, the same radius gives $1/x < -M$. There is consequently no two-sided real limit or common infinite limit.

At $+\infty$, consider $g(x) = (2x + 1)/(x + 2)$ on its natural domain, the inputs where its displayed formula is real and defined. The exact error from 2 is $3/|x + 2|$. Given $\varepsilon > 0$, choose $R > \max\{1, 3/\varepsilon\}$. For every $x > R$, the denominator is positive and the error is less than $3/x < \varepsilon$. Thus $g(x) \to 2$. This proof uses a large-input cutoff rather than a small-distance radius.

## 5.4 Continuity and its domain

**Definition 5.8** (Relative continuity)**.** For $f : A \to \mathbb{R}$ and $c \in A$, continuity at $c$ means

$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \forall x \in A : \quad |x - c| < \delta \implies |f(x) - f(c)| < \varepsilon.$$

The function is continuous on $A$ if it is continuous at every point of $A$.

The input $x = c$ is now allowed and has error zero. At a cluster point in the domain, continuity is exactly equality of the deleted limit with $f(c)$. At an isolated domain point, choose a radius containing no other domain point. The function is automatically continuous there. At the endpoint of a closed interval, only the available one-sided inputs are tested. Continuity does not require inputs outside the stated domain.

**Proposition 5.9** (Sequential continuity)**.** *A function $f : A \to \mathbb{R}$ is continuous at $c \in A$ exactly when every sequence $x_n \in A$ converging to $c$ satisfies $f(x_n) \to f(c)$.*

*Proof.* The forward implication follows by choosing the continuity radius for a requested error, then an input-sequence cutoff for that radius. Conversely, failure of continuity supplies one $\varepsilon_0 > 0$ and, for each $n$, an $x_n \in A$ with $|x_n - c| < 1/n$ and $|f(x_n) - f(c)| \geq \varepsilon_0$. These inputs tend to $c$, while their images do not tend to $f(c)$, contradicting the stated property. This proof also covers isolated points. There the proposed failure cannot supply any such input. ∎

Constants and the identity function are continuous: their errors are, respectively, zero and $|x - c|$. The algebraic laws therefore prove continuity of all polynomials and of rational functions wherever their denominators do not vanish. If $g$ is continuous at $c$, $f$ is continuous at $g(c)$ and $g$ maps its domain into the domain of $f$, then $f \circ g$ is continuous at $c$. Indeed, an input sequence approaching $c$ has inner images approaching $g(c)$. Outer continuity then gives the required composite image limit. This remains valid when some inner images equal $g(c)$.

Standing fact E assumes that the school exponential, logarithmic and trigonometric functions are continuous on their real domains. This uses the relative definition, including allowed domain endpoints. For a fixed $r \in \mathbb{R}$, continuity of $x^r$ on $x > 0$ follows from $x^r = \exp(r \ln x)$ and continuous composition. For $r > 0$, define $0^r = 0$. This gives continuity also at zero from the nonnegative domain: given $\varepsilon > 0$, take $\delta = \varepsilon^{1/r}$. Order of logarithms and exponentials shows that $0 < x < \delta$ implies $0 < x^r < \delta^r = \varepsilon$. The input zero has error zero. Thus this endpoint continuity is proved, rather than added to the standing assumption.

A *removable discontinuity* at a domain cluster point has a finite limit different from its assigned value. For instance, assigning the previous cancelled quotient the value 100 at 1

gives limit 2 but failure of continuity. Reassigning the value 2 repairs it. A *jump discontinuity* has two distinct finite one-sided limits. The function equal to $-1$ for $x < 0$ and 1 for $x \geq 0$ has a jump at zero. An *infinite discontinuity* has an infinite limit from at least one available side. The function $1/x$, assigned any real value at zero, is an example. Oscillation can prevent both finite and infinite limits, as $\sin(1/x)$ at zero shows. At a cluster point absent from a function's domain, assigning a value there is an extension problem. A continuous extension is possible exactly when a finite deleted limit exists there.

A continuous extension at a missing cluster point is unique when it exists. If the deleted limit is $L$, assign that value at the point. The existing radius controls all other inputs and the new point has error zero, proving continuity. Conversely, continuity would make the assigned value equal to the unique deleted limit.

**Example 5.10** (Two junction conditions)**.** Let $a, b \in \mathbb{R}$ and consider the real function

$$f(x) = \begin{cases} ax + b, & x < 0, \\ x^2 + 1, & 0 \leq x \leq 1, \\ 3x - a, & x > 1. \end{cases}$$

Each piece is continuous in its own interval. At zero the assigned value is 1 and the right limit is 1, while the left limit is $b$. Thus continuity at zero requires $b = 1$. At 1 the assigned value and left limit are 2, while the right limit is $3 - a$. Thus continuity there requires $a = 1$. These conditions are also sufficient: the pieces are continuous away from the junctions, and both one-sided limits at each junction equal its assigned value. Consequently $f$ is continuous on all of $\mathbb{R}$ exactly when $a = b = 1$. Testing just one junction would leave an undetermined parameter.

**Example 5.11** (A bounded function continuous nowhere)**.** Define $D : \mathbb{R} \to \mathbb{R}$ by $D(x) = 1$ for rational $x$ and $D(x) = 0$ otherwise. Every open interval contains a rational number by Lecture 2. It also contains an irrational number. For an interval $(u, v)$, choose rational $q$ between $u - \sqrt{2}$ and $v - \sqrt{2}$. Then $q + \sqrt{2}$ lies in $(u, v)$ and is irrational: rationality would imply rationality of $\sqrt{2}$ after subtracting $q$.

Fix any $c \in \mathbb{R}$ and set $\varepsilon_0 = 1/2$. If $c$ is rational, every proposed radius contains an irrational input, whose value differs from $D(c)$ by 1. If $c$ is irrational, every radius contains a rational input with the same unit error. Thus the continuity condition fails at every point. The bound $0 \leq D \leq 1$ does not supply continuity. The domain contains both kinds of inputs arbitrarily near every point, so restricting attention to one kind would miss the failure.

## 5.5 The basic sine limit

**Theorem 5.12** (Sine limit in radians)**.** *As $x \to 0$, $\sin x/x \to 1$.*

*Proof.* Standing fact E gives $\cos x \leq \sin x/x \leq 1$ when $0 < |x| < \pi/2$. Its continuity clause gives $\cos x \to \cos 0 = 1$. The function squeeze theorem therefore proves the result. ∎

For example, for $x \neq 0$,

$$\frac{\sin(5x)}{2x} = \frac{5}{2}\frac{\sin(5x)}{5x} \longrightarrow \frac{5}{2}.$$

The inner argument tends to zero and avoids zero, so the composition hypothesis is met. The function $S(x) = \sin x/x$ for $x \neq 0$, with $S(0) = 1$, is continuous at zero by the proved limit. It is continuous elsewhere by the quotient law, so this extension is continuous on $\mathbb{R}$.

**Summary 5.**

- Definition 5.1 and Theorem 5.3: deleted neighbourhood limits at cluster points are equivalent to limits along every allowed approaching sequence.
- Theorems 5.5 and 5.7: arithmetic, order and squeeze laws hold. Composition must handle inner values equal to the outer point.
- One-sided and infinite limits change the input tests or output bounds, with the same order of tolerance and cutoff choices.
- Definition 5.8 and Proposition 5.9: continuity is relative to the domain and includes the actual value at the point.
- Theorem 5.12: $\sin x/x \to 1$ in radians, using the stated school bounds and continuity assumption E.

Lecture 6 uses these tools to prove logarithmic and exponential standard limits and the intermediate value theorem. Sheet 5 practises distance definitions, sequential tests and continuity.

## References

R. G. Bartle and D. R. Sherbert, *Introduction to Real Analysis*, 4th ed., Wiley, 2011, Chapters 4–5.